WEEKLY PROGRESS UPDATE

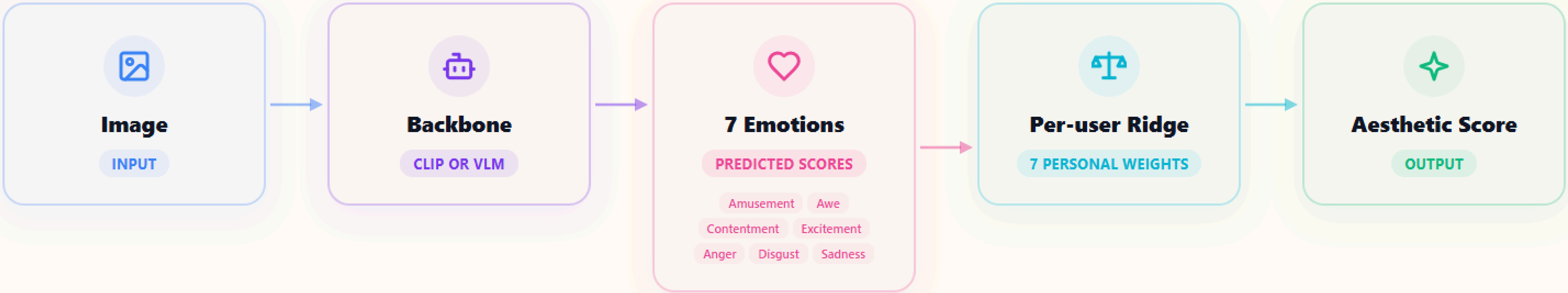
Emotion-Mediated Personalized Image Aesthetic Assessment

Predicting individual aesthetic preferences through emotional responses

Presenter: Pinwa

02 — THE PIPELINE

How it works.



The emotion layer in the middle makes the whole thing **interpretable**.

03 — QUICK RECAP

Last two weeks' Key findings.

WEEK 1

Emotion mediation is feasible

P-oracle = 0.72 (upper bound). When we feed in the true emotions, we clearly beat predicting beauty directly.

WEEK 2A

Emotion formula vs personality

No strong link found. The beauty formula does not reduce to Big Five personality traits in any simple way.

WEEK 2B

Formula stable across image types

The personal emotion formula stays consistent across Art, Fashion, and Landscape domains. New finding.

04 — THIS WEEK

Three things I did.

1

A fair comparison

I re-ran Hayashi-san's ICI and MIR baselines on exactly the same data split as ours, so the comparison is fair.

2

A different backbone

I tried a vision-language model, Qwen3-VL-8B, in place of CLIP, following the Ryu & Yanaka paper Hayashi-san recommended.

3

Fine-tuning the backbone

Instead of keeping the backbone frozen, I tried fine-tuning it to predict emotions.

05 — TASK 1

Making the comparison fair.

BEFORE

Random split, seed=42

Same **COUNT** (100/50) but different **IMAGES** than ICI/MIR.

AFTER

Read train_PIAA.txt / test_PIAA.txt directly

Same images as ICI/MIR. Also found & fixed duplicate ratings (test-retest) → averaged them.

TABLE 1 — FAIR COMPARISON, CCC BY DOMAIN (N_TRAIN=100)

Model	art	fashion	landscape	avg
P-oracle (ceiling)	0.725	0.715	0.734	**0.725**
ICI (Hayashi-san)	0.480	0.315	0.430	**0.409**
MIR (Hayashi-san)	0.484	0.313	0.435	**0.411**
Ours: CLIP Hybrid	0.456	0.264	0.377	**0.366**
Ours: CLIP Direct	0.384	0.202	0.292	**0.293**

06 — TASK 1 — RESULT

Does emotion mediation really help?

CLIP

N	MEDIAN Δ	HYBRID WIN %	EFFECT R
10	+0.007	68.0%	0.36
25	+0.057	84.0%	0.70
50	+0.067	89.9%	0.79
100	+0.071	91.7%	0.82

Qwen2-vision

N	MEDIAN Δ	HYBRID WIN %	EFFECT R
10	+0.001	68.0%	0.38
25	+0.053	78.3%	0.62
50	+0.058	79.3%	0.64
100	+0.049	80.1%	0.62

What this tells us

With only 10 images per person, the emotion step barely helps. But from 25 images up, it helps a lot, and stays strong. We see the same shape for both backbones, so this isn't a quirk of one model. Every result here is statistically significant (Wilcoxon signed-rank test, $p < 0.001$).

07 — TASK 2

Trying a VLM backbone: Qwen3-VL-8B.

REFERENCE

Ryu & Yanaka Configuration

- Qwen3-VL model
- Layer 15, using the text tokens
- Average pooling
- A specific prompt
- StandardScaler before Ridge

OUR APPROACH

My Implementation

- Used the largest 8B model, not the 4B they recommend
- Saved every layer in a single GPU run
- So I can pick the best layer later, without borrowing the GPU again

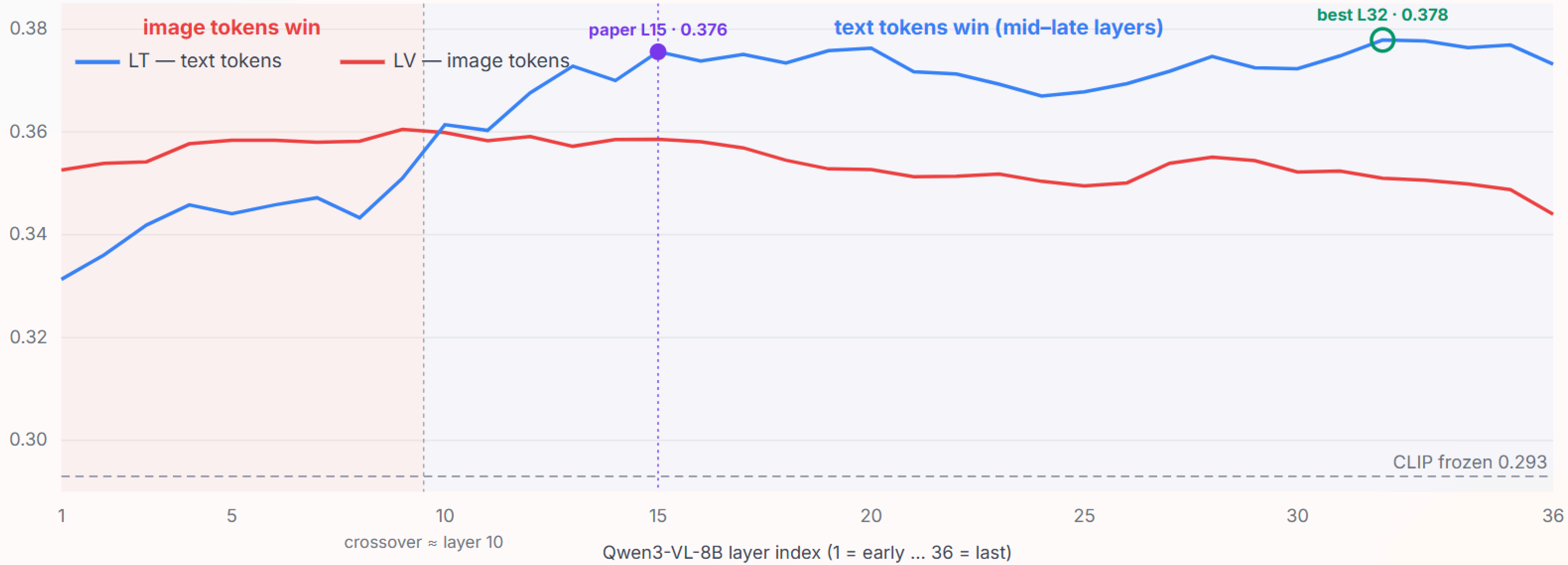
Why StandardScaler matters.

Backbone	Preprocessing	CCC ↑	SROCC ↑
Qwen3-VL-8B (4096d)	L2-norm	0.006	0.385
Qwen3-VL-8B (4096d)	StandardScaler	0.376	0.413
CLIP (512d)	L2-norm	0.301	0.413
CLIP (512d)	StandardScaler	0.293	0.399

Notice that CLIP, with only 512 numbers per image, barely changes. Only the VLM with 4096 numbers collapses. So the problem comes from having so many dimensions, and StandardScaler is what fixes it.

Once the scaling was fixed, the VLM did beat CLIP, by about 0.08 in CCC. And from the middle layers onward the text tokens worked better than the image tokens — the best was a mid-to-late text-token layer, matching the paper's choice of text tokens.

Which layer, and text vs image tokens.

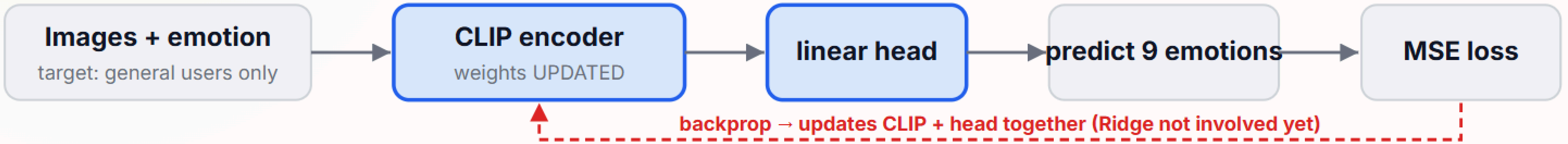


As the model reads deeper, the image information moves from the image tokens into the text tokens. So the best features are text tokens at a middle-to-late layer. Layer 15 (the paper's choice) is within 0.002 of my overall best — effectively tied.

10 — TASK 3 — HOW FINE-TUNING WORKS

Fine-tune, then freeze, then personalize.

Stage A — fine-tune (population, per fold)



↓ training done → FREEZE the fine-tuned CLIP (weights never change again)

Stage B — freeze + personalize (per user, no backprop into CLIP)

Same idea as the baselines: pretrain the backbone on the population, then personalize on top — not one joint model.

Why two stages, not one joint model? Each user has only ~100 images — fine-tuning CLIP per user would overfit. Keeping a fixed backbone plus a small linear per-user formula stays data-efficient and interpretable.

11 — TASK 3 — RESULTS

Fine-tuning & Main Comparison.

Backbone	Direct CCC	Hybrid CCC	How much the emotion step helps
CLIP frozen	0.293	0.366	+0.073
★ CLIP-ft (emotion)	0.386	0.400	+0.014
CLIP-ft (overall)	0.368	0.387	+0.019
Qwen2-vision	0.321	0.357	+0.036
Qwen3-VL-8B	0.376	0.382	+0.006
ICI (Hayashi-san)	–	0.409	–
MIR (Hayashi-san)	–	0.411	–
P-oracle (ceiling)	–	*0.725*	–

A small fine-tuned **CLIP (0.400)** does better than the much larger frozen Qwen3 (0.382). We don't need a huge model to get good results.

A pattern worth noting: the stronger the backbone, the less the emotion step adds (`+0.073 → +0.006`). A good backbone already captures emotion on its own.

These numbers are leak-free: the people we test on are never used to build the training target.

12 — SUMMARY

Key findings this round.

1 The comparison is now fair

Same data split, and duplicate ratings averaged. We now compare with ICI and MIR on exactly the same images.

2 The emotion step really helps

The Wilcoxon test shows a significant improvement for both backbones ($p < 0.001$), so it isn't down to chance.

3 It helps once you have 25+ images

With fewer than 25 images per person, the emotion step barely helps. Above that, it clearly does.

4 Scaling matters for big features

The 4096-number VLM features collapse without StandardScaler. The 512-number CLIP features are barely affected.

5 A small tuned model can win

Fine-tuned CLIP (0.400) does better than the frozen Qwen3-VL-8B (0.382). We don't need a huge model.

6 Better backbone, less to add

The stronger the backbone, the less the emotion step adds, because a good backbone already carries emotion.

13 — NEXT STEPS

What comes next.

What I plan to try next

1. **Cold-start analysis**
   How many ratings needed before personal model beats population formula?
2. **Noise-ceiling analysis**
   ~4,500 test-retest cases. Estimate human self-consistency → true upper bound for any model.
3. **Trait-conditioned emotion prediction**
   Can trait vectors predict how a user's emotions deviate from population? If yes → zero-shot personalization.

Questions I'd like to discuss

1. Is it worth fine-tuning Qwen too, or is "small fine-tuned CLIP beats giant frozen Qwen" strong enough?
2. Should I keep chasing accuracy, or start focusing on interpretability findings?
3. Can I use my reproduced ICI/MIR numbers (0.409/0.411) as the official comparison, since they're on exactly the same split?
4. Given these findings, is IEEE Access still the right venue? Related work to check?

Not in a hurry — can continue from Thailand after the internship if needed.

Thank You!

Happy to take any questions or suggestions.